

1. STEM and Societal Impact

STEM plays an important role in improving society because it helps solve real-world problems through innovation and technology. From education and healthcare to communication and automation, STEM creates opportunities that can positively impact communities.

What motivates me to teach STEM is the ability to simplify concepts and help students build confidence in problem-solving. During my academic projects and while helping peers with programming concepts, I realised that many students learn better when concepts are connected to real-life examples.

I believe STEM education becomes more meaningful when students are encouraged to think practically, ask questions, and apply their learning to everyday problems.

2. Making CS and AI Education More Equitable

I believe CS and AI education needs to become more practical and accessible and less intimidating for students from underserved communities. Many students lose interest because concepts are introduced in a very theoretical or complex way.

From my experience helping peers and juniors with programming, I noticed students understand concepts much faster when topics are explained with relatable examples instead of focusing only on syntax.

I think project-based learning, interactive teaching, and real-world applications can make CS and AI education more meaningful and future-ready for students from all backgrounds.

4. If Only 30% Completed the Pre-work

Before the session, I would send a quick reminder along with a short recap resource to help students prepare at least partially.

During the session, I would begin with a brief interactive revision of if-else logic using simple real-life examples so students who missed the pre-work can catch up. At the same time, I would keep prepared students engaged by asking them to solve small challenge questions or explain concepts to others.

My focus would be on maintaining participation and ensuring no student feels left behind while still keeping the session productive.

5. Tracking Student Progress

I would use Google Sheets because it is simple, collaborative, and supports real-time tracking and dashboards.

Key metrics I would track:

- Attendance and participation
- Assignment/pre-work completion
- Quiz or assessment scores
- Improvement over time

These metrics would help identify students who need additional support and also help measure the effectiveness of teaching methods.